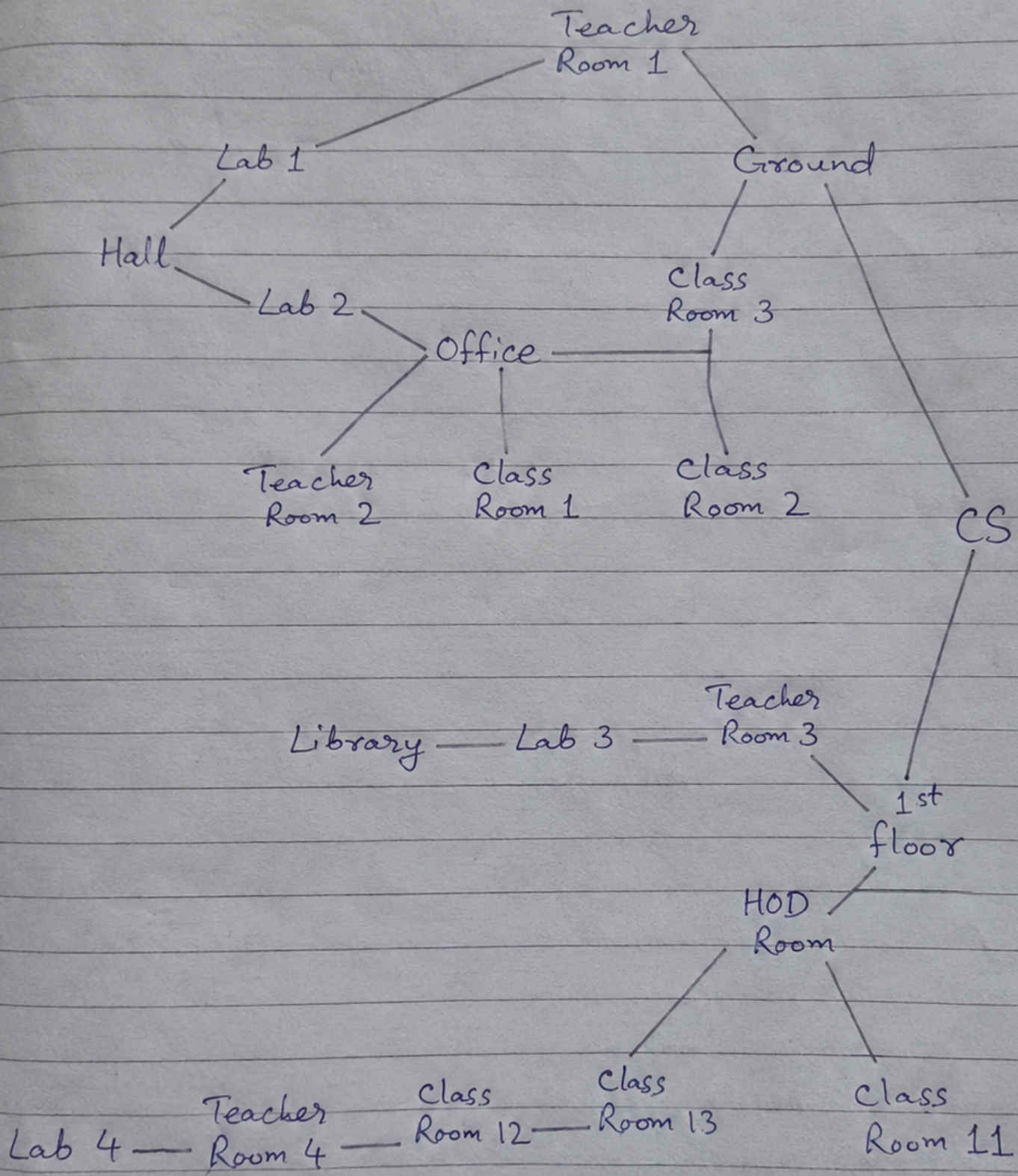
Graph

Date:


Teacher Room 1
Lab 1
Ground
Hall
Lab 2
Class Room 3
Office
Teacher Room 2
Class Room 1
Class Room 2
CS
Library
Lab 3
Teacher Room 3
1st floor
HOD Room
Lab 4
Teacher Room 4
Class Room 12
Class Room 13
Class Room 11

Date:

Transition Table :

CS	→	Ground , 1st floor
Ground	→	Teacherroom1 , Classroom3 , Classroom2 , CS
Teacher Room 1	→	Ground , Lab 1
Classroom 3	→	Ground , Office
Classroom 2	→	Ground , classroom 1
Lab 1	→	Teacherroom 1 , Hall
Office	→	Classroom 3 , Lab 2 , Teacher room 2
Classroom 1	→	Classroom 2 , Teacherroom 2
Lab 2	→	Office , Hall
Teacher room 2	→	Office , Classroom 1
Hall	→	Lab 1 , Lab 2
1st floor	→	Teacherroom 3 , HOD room , Classroom 11
Teacherroom 3	→	1st floor , Lab 3
HOD room	→	1st floor , Classroom 13
Classroom 11	→	1st floor , Classroom 13
Lab 3	→	Teacher room 3 , Library
Classroom 13	→	HODroom , Classroom 13 , Classroom 11
Classroom 12	→	Classroom 13 , Teacher room 4
Teacher room 4	→	Classroom 12 , Lab 4
Lab 4	→	Teacher room 4
Library	→	Lab 3

Date:

Formulation of the Problem:

Initial State :
Library

Goal State :
Classroom 13

State space :
All accessible location in the CS department.

Action :
Move from one accessible state to an other adjacent state

Transition model :
Purpose to move state

Path cost :
Distance.

Date:

BFS TABLE:

Initialization:
Q: [Library]
V: [Library]

deque Library & Expand
Q: [Lab 3]
V: [Library, Lab 3]

deque Lab 3 & Expand
Q: [TR 3]
V: [Library, Lab 3, TR 3]

deque TR 3 & Expand
Q: [1st floor]
V: [Library, Lab 3, TR 3, 1st floor]

deque 1st floor & Expand
Q: [CS, HOD, CR 11]
V: [Library, Lab 3, TR 3, 1st floor, CS, HOD, CR 11]

Date:

deque CS & Expand
Q: [HOD , CR 11, Ground]

V: [Library , Lab 3 , TR 3 , 1st floor , CS ,
HOD , CR 11, Ground]

deque HOD & Expand
Q: [CR 11 , Ground, CR 13]

V: [Library , Lab 3 , TR 3 , 1st floor , CS ,
HOD , CR 11 , Ground , CR 13]

Date:_____

Shortest Path (Library - CR 13)

